

CS736 - Assignment-1

Team Members

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Under the Guidance of :

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Q1 - Optimal Parameters

Prior: Quadratic

1. $\alpha^* = 0.1062$
2. $\text{RRMSE}(\alpha^*) = 0.28108$
3. $\text{RRMSE}(1.2\alpha^*) = 0.28141$
4. $\text{RRMSE}(0.8\alpha^*) = 0.28153$

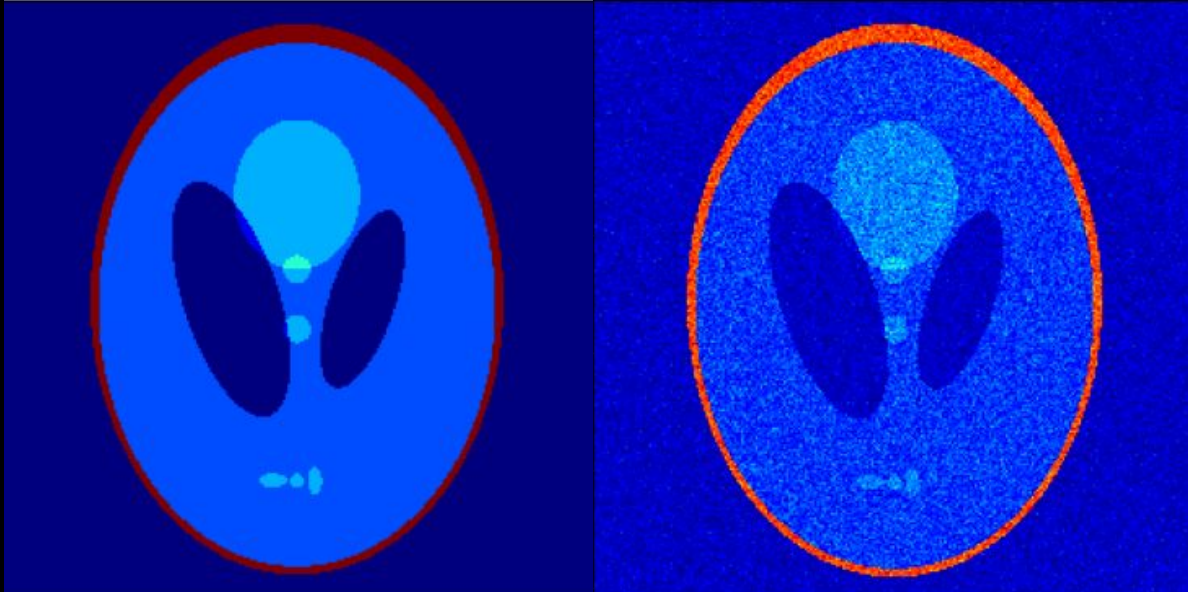
Prior: Discontinuity-adaptive Huber

1. $\alpha^* = 0.96612,$
 $\gamma^* = 0.04165$
2. $\text{RRMSE}(\alpha^*, \gamma^*) = 0.24063$
3. $\text{RRMSE}(1.2\alpha^*, \gamma^*) = 0.24063$
4. $\text{RRMSE}(0.8\alpha^*, \gamma^*) = 0.24281$
5. $\text{RRMSE}(\alpha^*, 1.2\gamma^*) = 0.24110$
6. $\text{RRMSE}(\alpha^*, 0.8\gamma^*) = 0.24075$

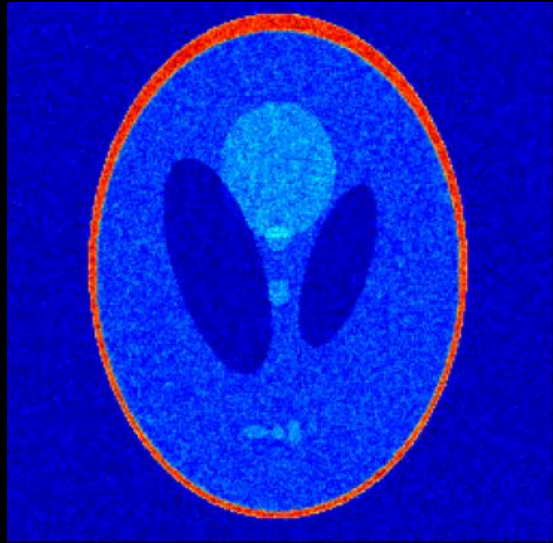
Prior: Discontinuity-adaptive

1. $\alpha^* = 0.93340,$
 $\gamma^* = 0.04064$
2. $\text{RRMSE}(\alpha^*, \gamma^*) = 0.24492$
3. $\text{RRMSE}(1.2\alpha^*, \gamma^*) = 0.24492$
4. $\text{RRMSE}(0.8\alpha^*, \gamma^*) = 0.24952$
5. $\text{RRMSE}(\alpha^*, 1.2\gamma^*) = 0.24493$
6. $\text{RRMSE}(\alpha^*, 0.8\gamma^*) = 0.24552$

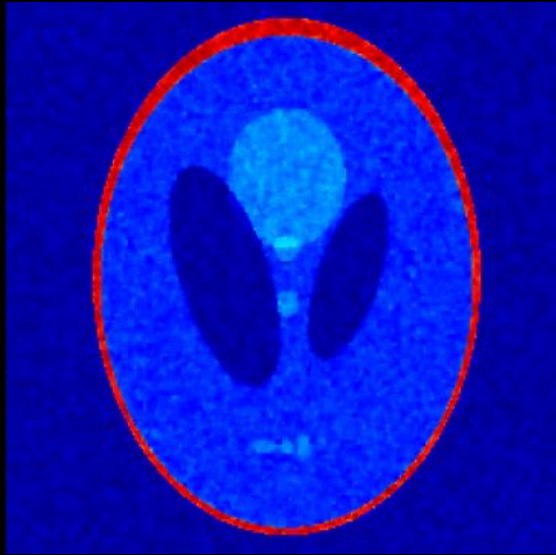
Q1 - Noiseless and Noisy Image



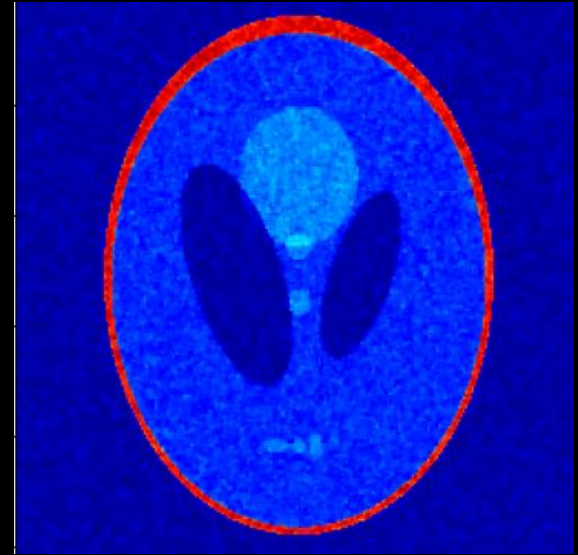
Q1 - Denoised Images



Quadratic

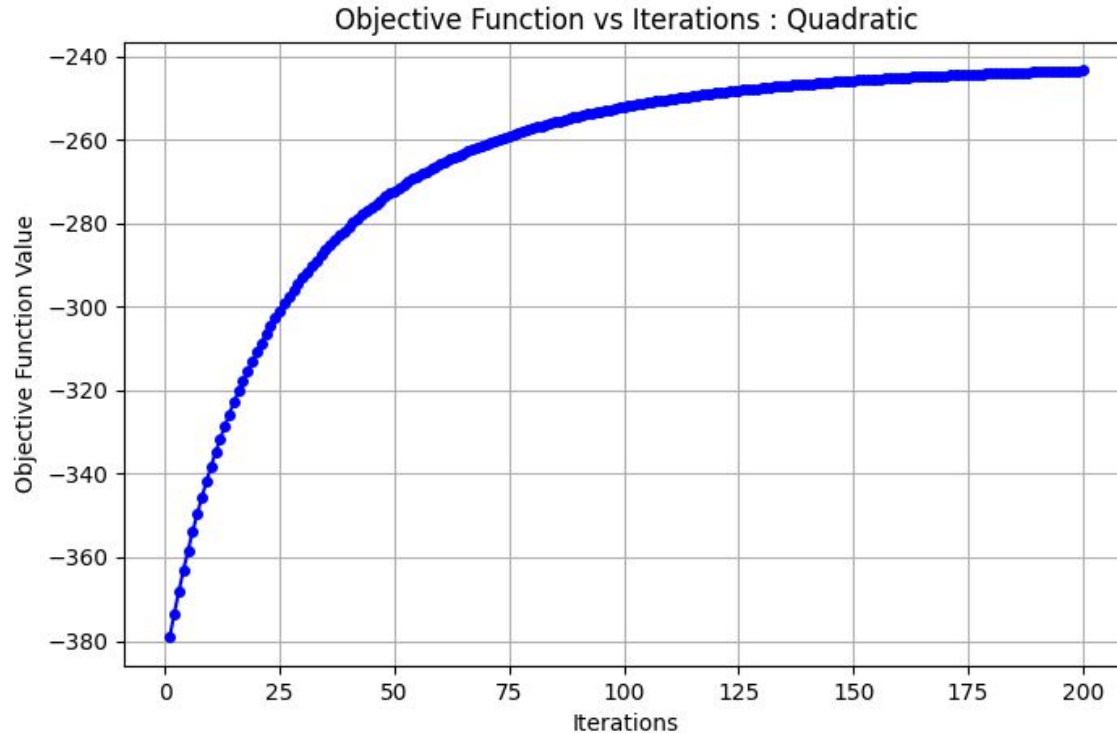


Discontinuity-adaptive
Huber

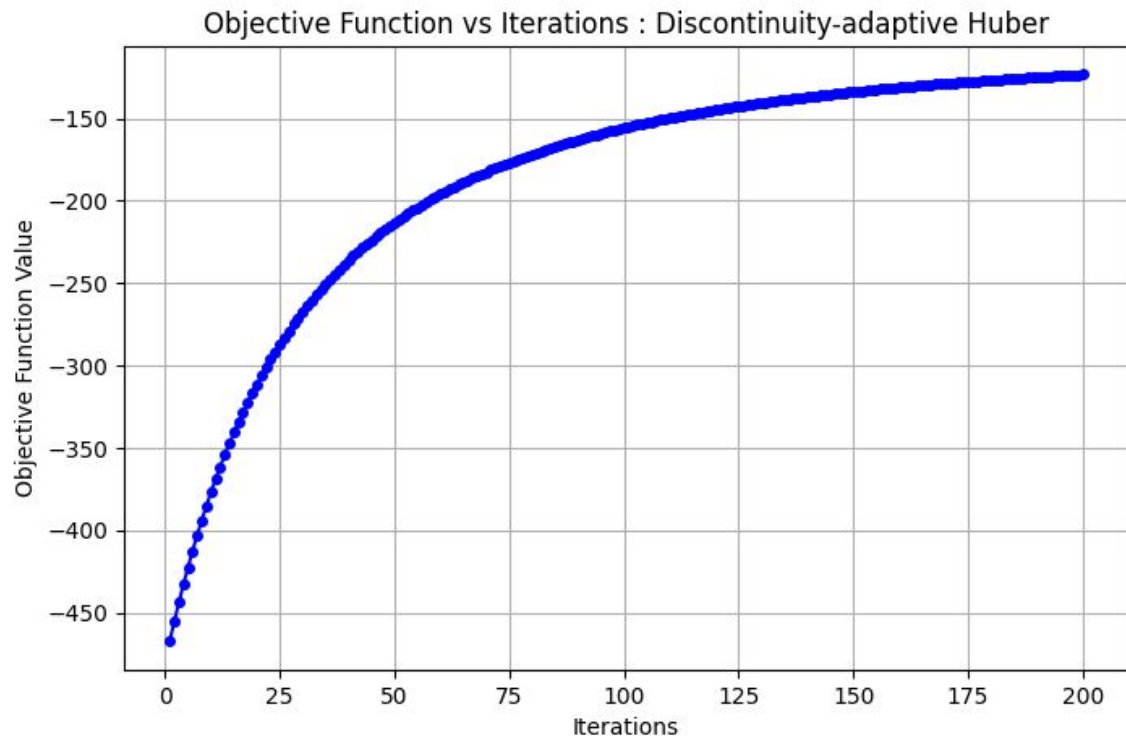


Discontinuity-adaptive

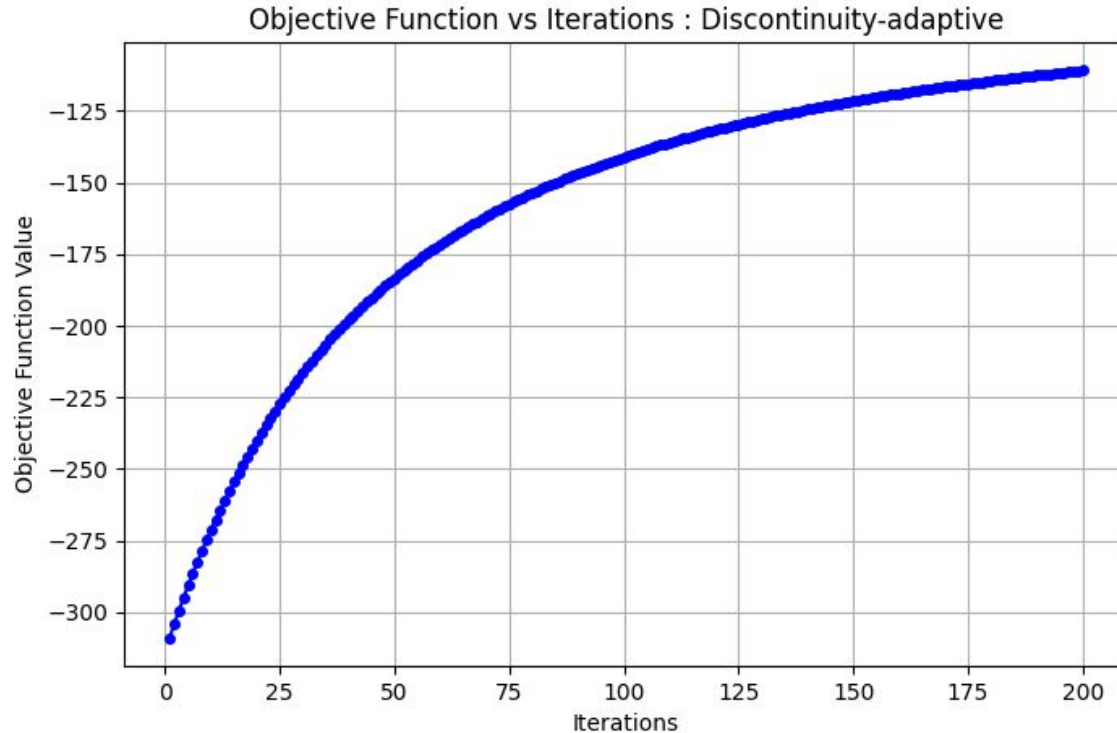
Q1 - Objective Function Plots



Q1 - Objective Function Plots



Q1 - Objective Function Plots



Q2 - Optimal Parameters

Prior: Quadratic

1. $\alpha^* = 0.14228$
2. $\text{RRMSE}(\alpha^*) = 0.12206$
3. $\text{RRMSE}(1.2\alpha^*) = 0.12246$
4. $\text{RRMSE}(0.8\alpha^*) = 0.12255$

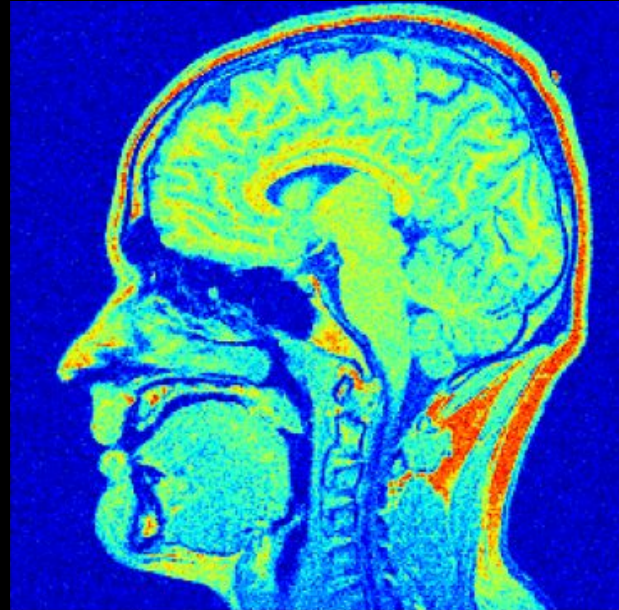
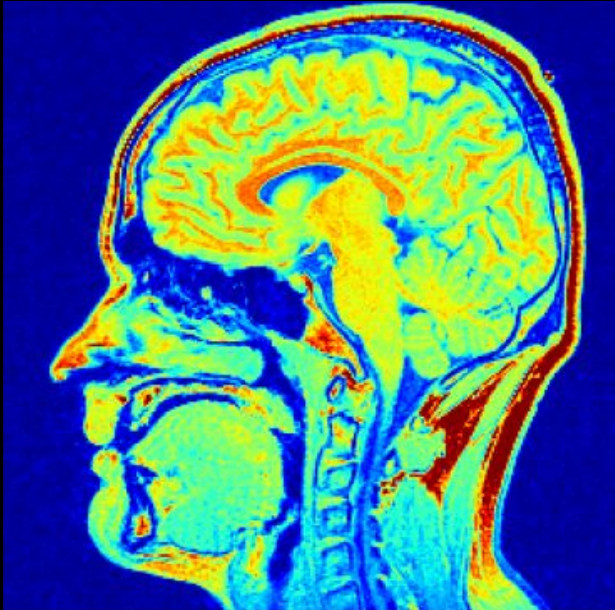
Prior: Discontinuity-adaptive Huber

1. $\alpha^* = 0.47034,$
 $\gamma^* = 0.06432$
2. $\text{RRMSE}(\alpha^*, \gamma^*) = 0.11421$
3. $\text{RRMSE}(1.2\alpha^*, \gamma^*) = 0.11446$
4. $\text{RRMSE}(0.8\alpha^*, \gamma^*) = 0.11602$
5. $\text{RRMSE}(\alpha^*, 1.2\gamma^*) = 0.11459$
6. $\text{RRMSE}(\alpha^*, 0.8\gamma^*) = 0.11458$

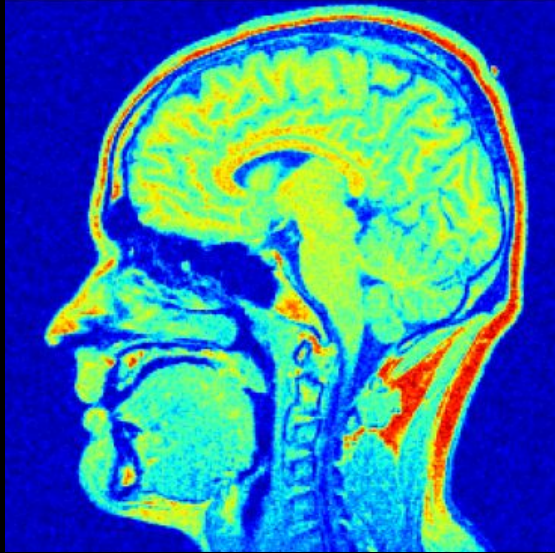
Prior: Discontinuity-adaptive

1. $\alpha^* = 0.56075,$
 $\gamma^* = 0.07235$
2. $\text{RRMSE}(\alpha^*, \gamma^*) = 0.11517$
3. $\text{RRMSE}(1.2\alpha^*, \gamma^*) = 0.11518$
4. $\text{RRMSE}(0.8\alpha^*, \gamma^*) = 0.11749$
5. $\text{RRMSE}(\alpha^*, 1.2\gamma^*) = 0.11528$
6. $\text{RRMSE}(\alpha^*, 0.8\gamma^*) = 0.11548$

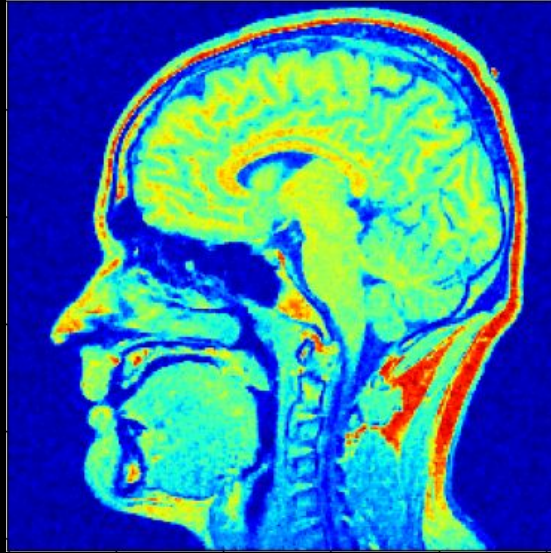
Q2 - Noiseless/Noisy Image



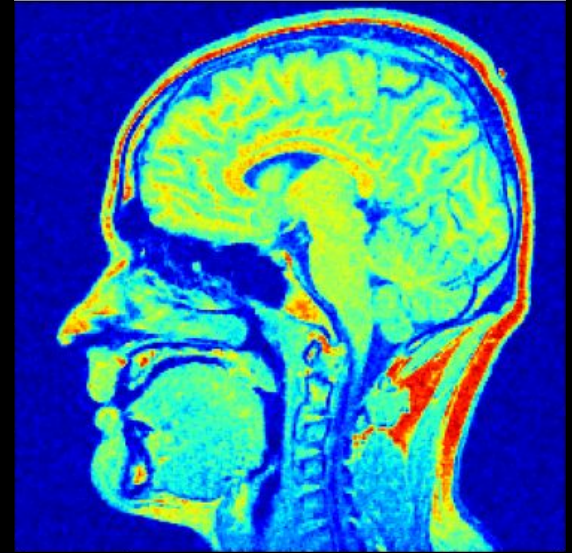
Q2 - Denoised Images



Quadratic

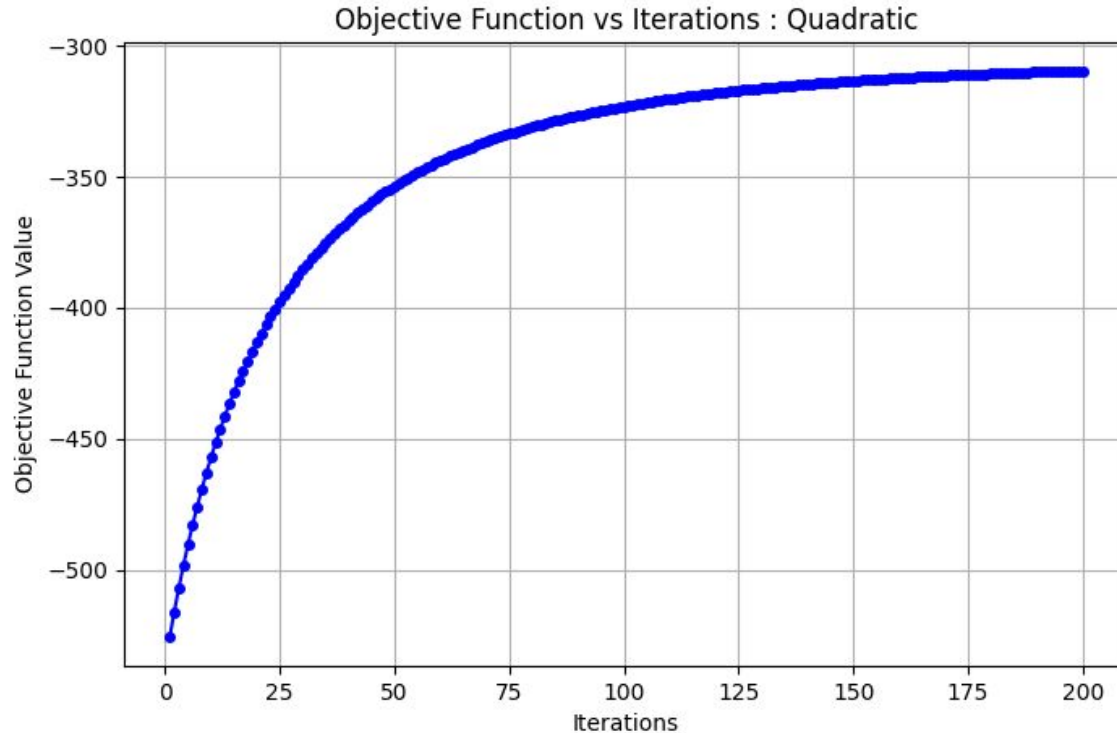


Discontinuity-adaptive
Huber

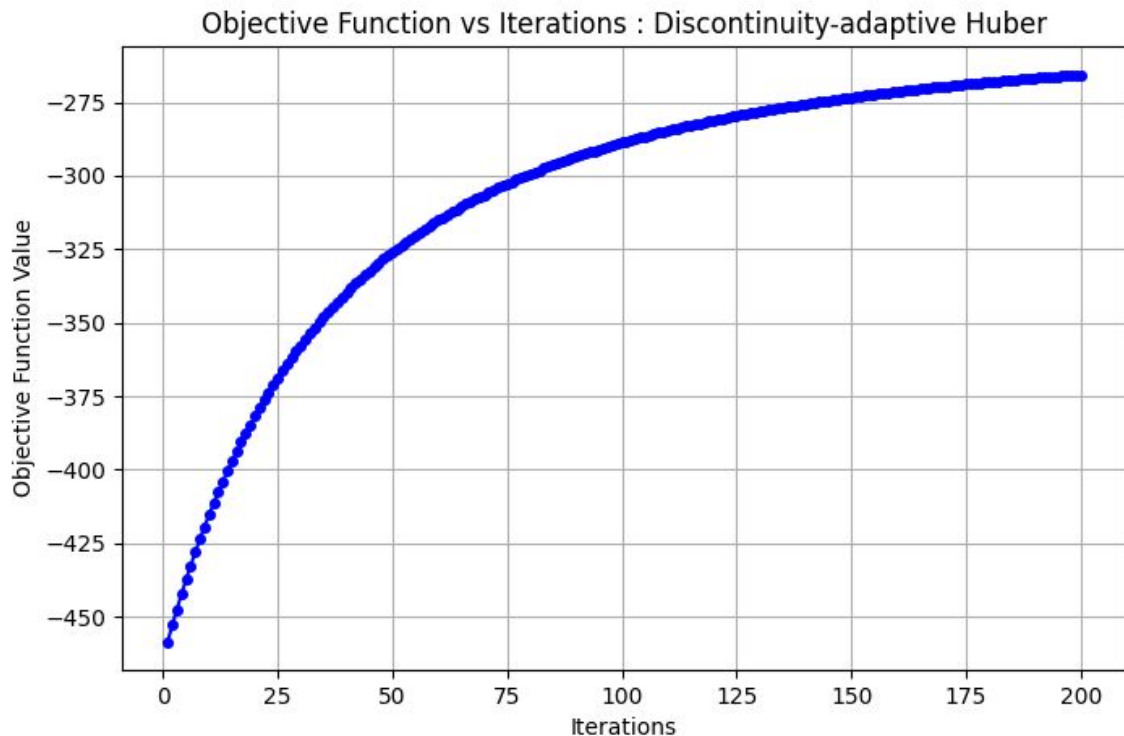


Discontinuity-adaptive

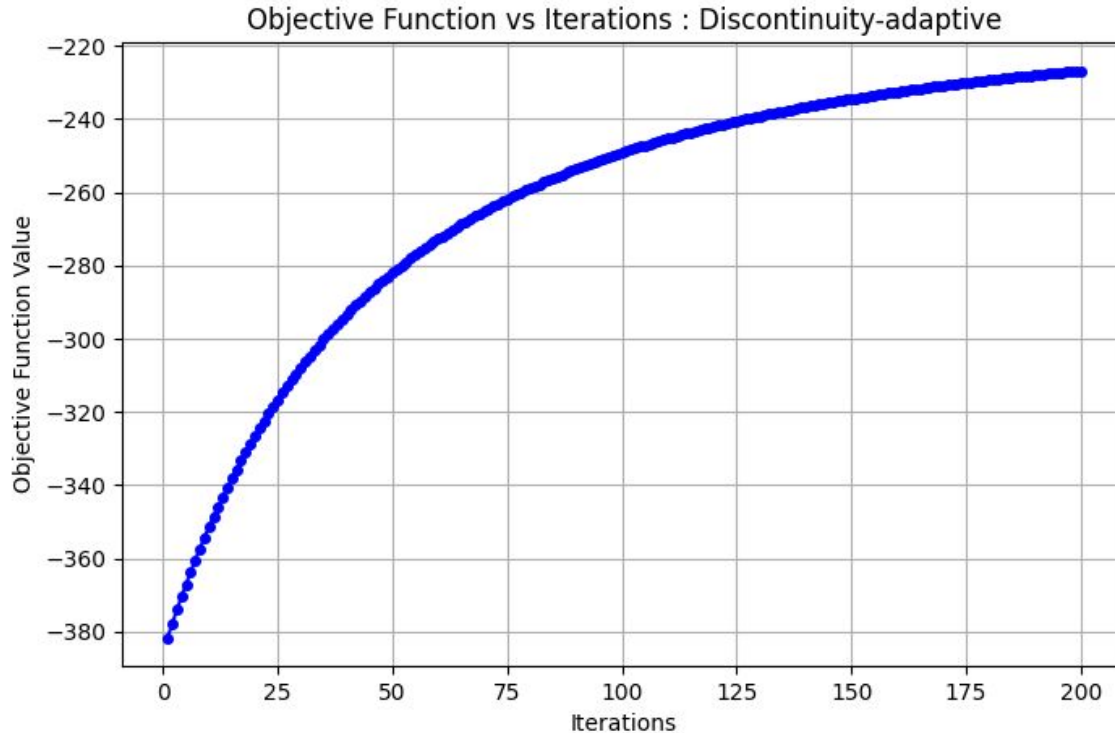
Q2 - Objective Function Plots



Q2 - Objective Function Plots



Q2 - Objective Function Plots



Q3 - Optimal Parameters

Prior: squared-L2

1. $\alpha^* = 0.5477$
2. $\text{RRMSE}(\alpha^*) = 0.17701$
3. $\text{RRMSE}(1.2\alpha^*) = 0.17712$
4. $\text{RRMSE}(0.8\alpha^*) = 0.17718$

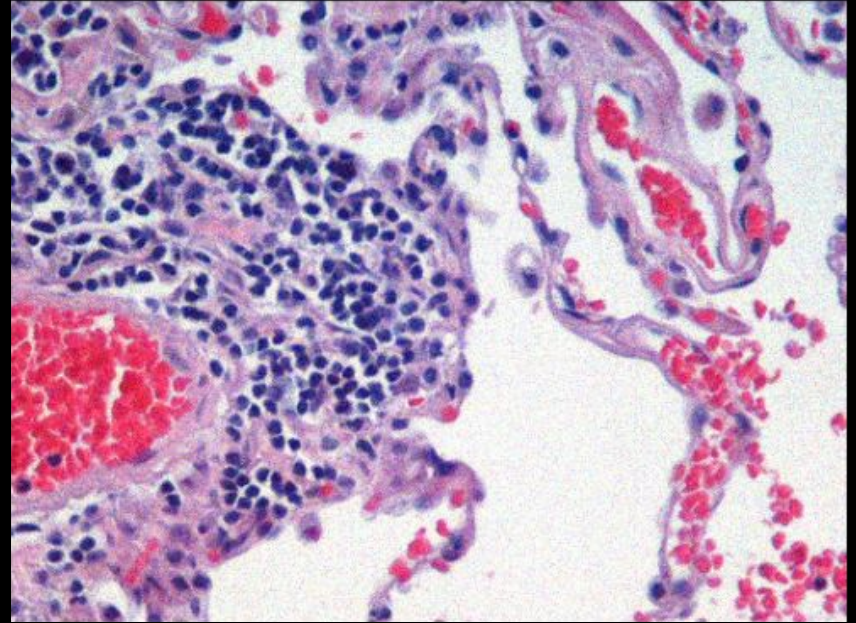
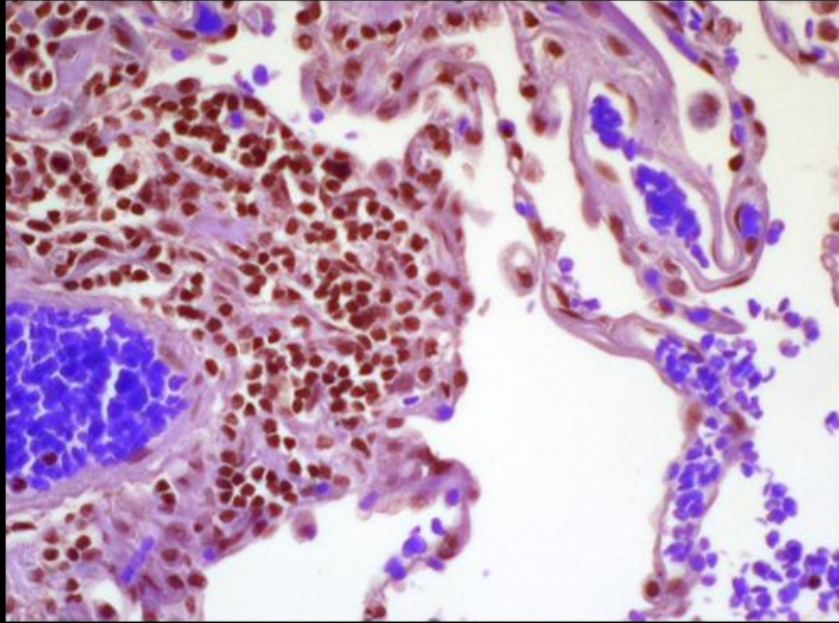
Prior: L2-norm

1. $\alpha^* = 0.0955$
2. $\text{RRMSE}(\alpha^*) = 0.17743$
3. $\text{RRMSE}(1.2\alpha^*) = 0.17760$
4. $\text{RRMSE}(0.8\alpha^*) = 0.17761$

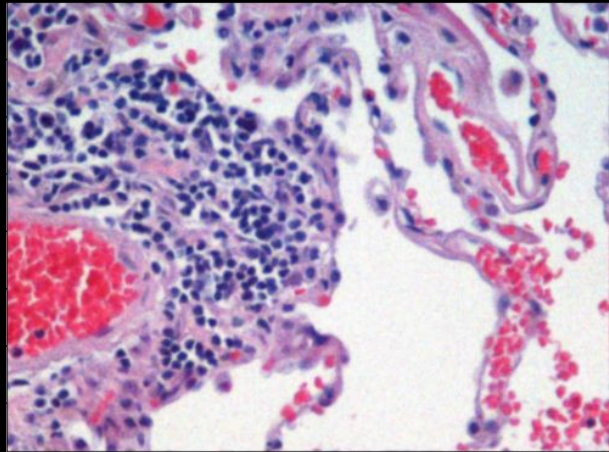
Prior: (Huber-regularized)

1. $\alpha^* = 0.4545, \gamma^* = 0.1558$
2. $\text{RRMSE}(\alpha^*, \gamma^*) = 0.17738$
3. $\text{RRMSE}(1.2\alpha^*, \gamma^*) = 0.17755$
4. $\text{RRMSE}(0.8\alpha^*, \gamma^*) = 0.17771$
5. $\text{RRMSE}(\alpha^*, 1.2\gamma^*) = 0.17748$
6. $\text{RRMSE}(\alpha^*, 0.8\gamma^*) = 0.17756$

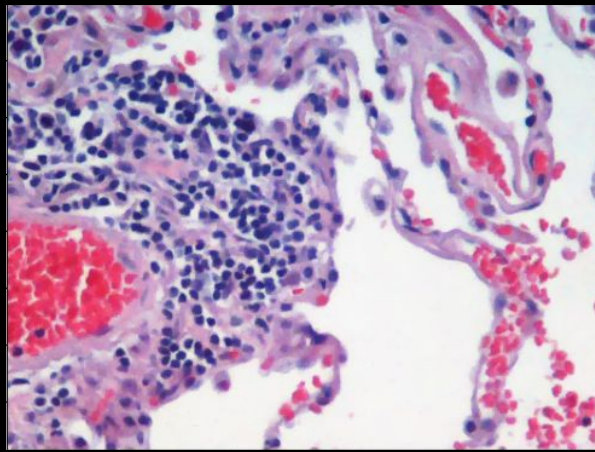
Q3 - Noiseless / Noisy Image



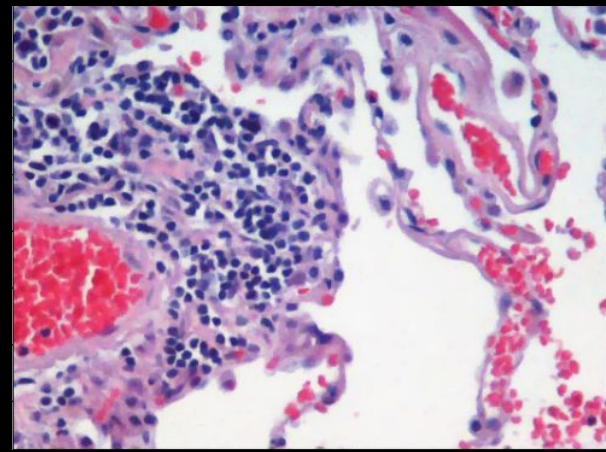
Q3 - Denoised Images



squared-L2

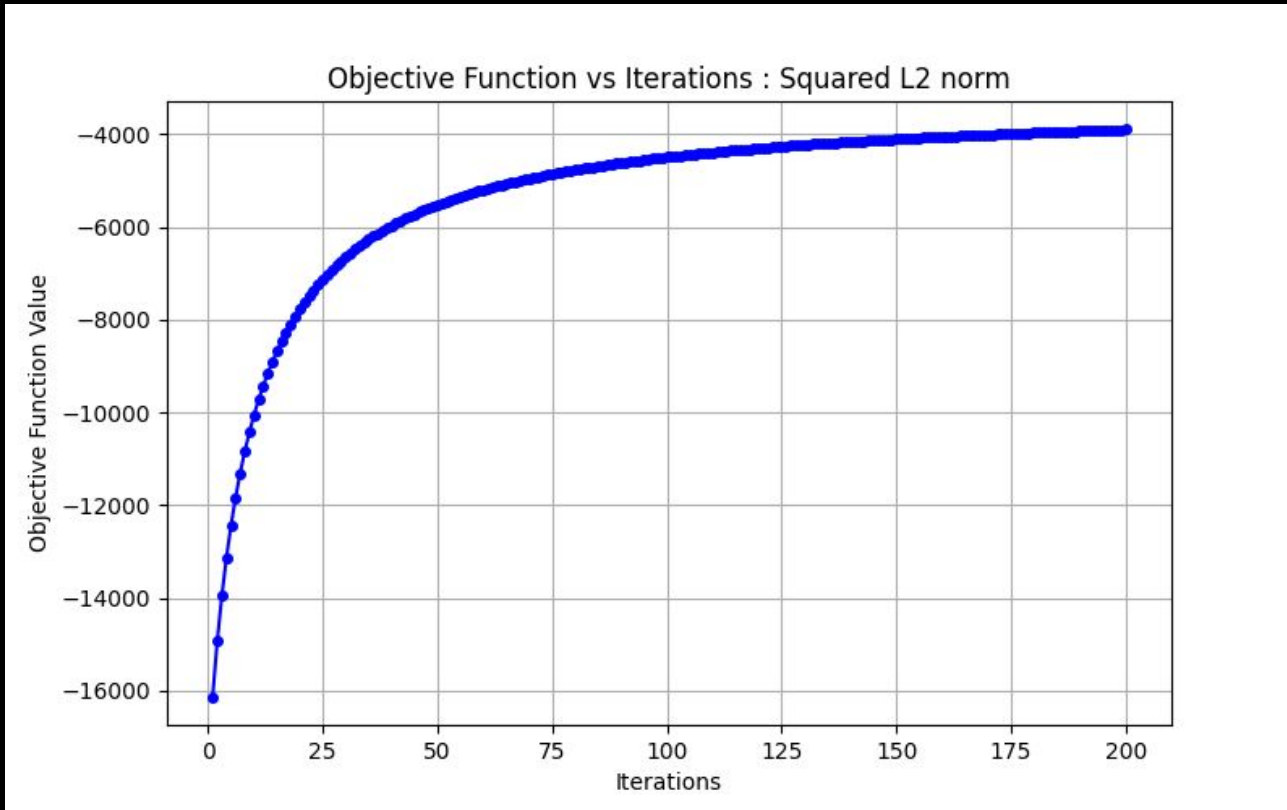


L2-Norm

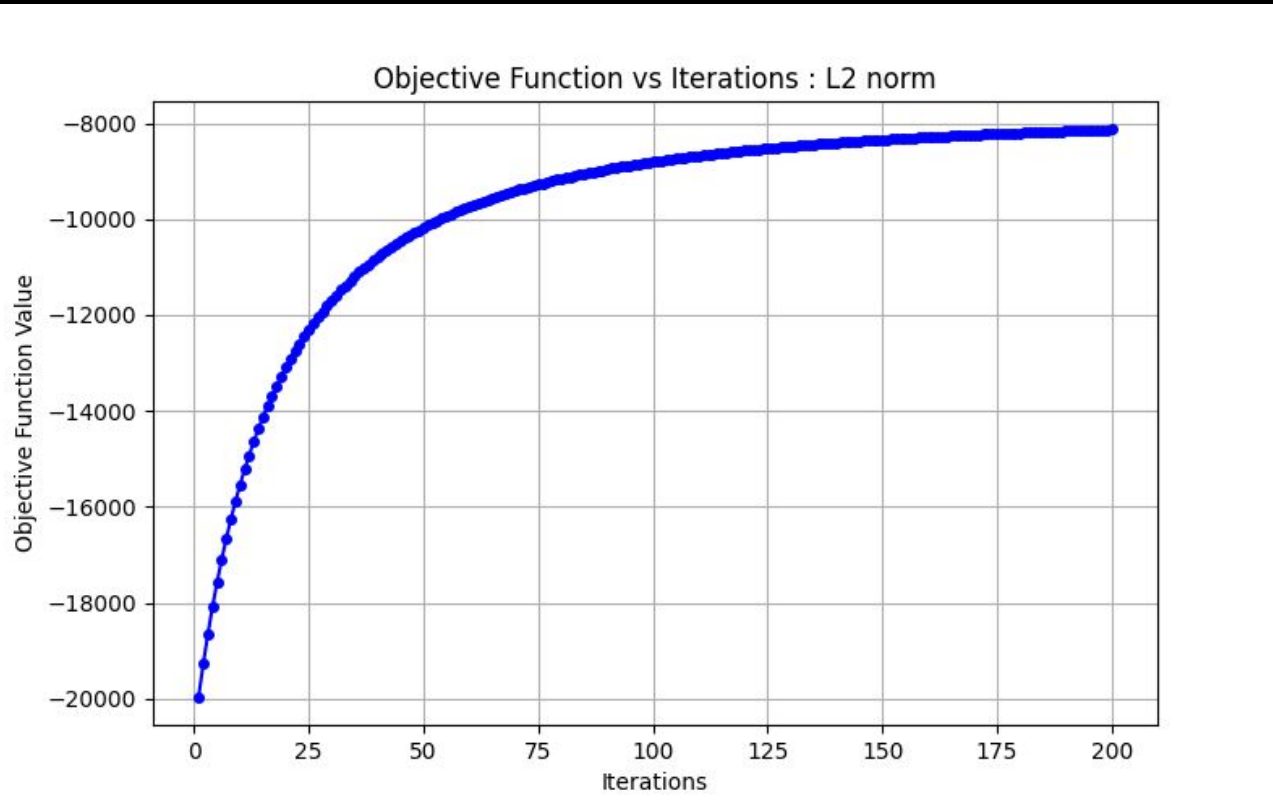


Huber-L1

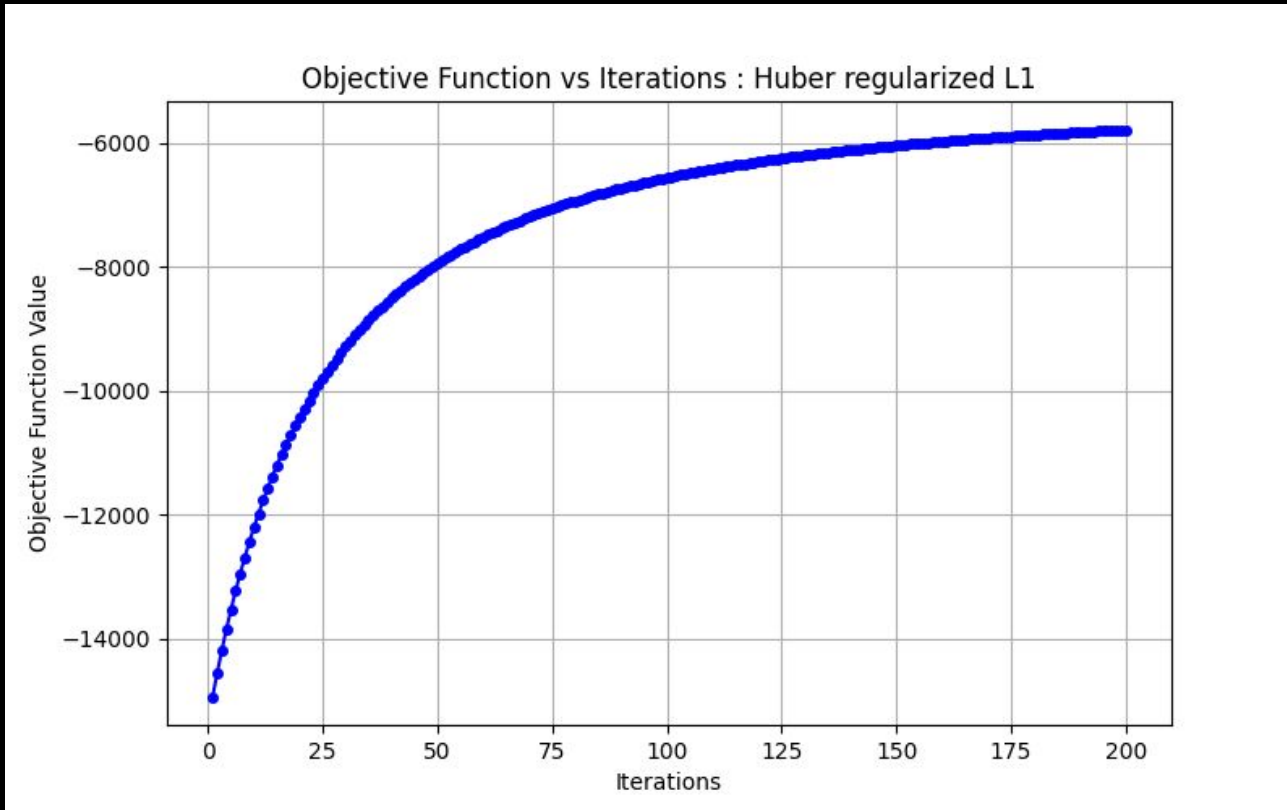
Q3 - Objective Function Plots



Q3 - Objective Function Plots



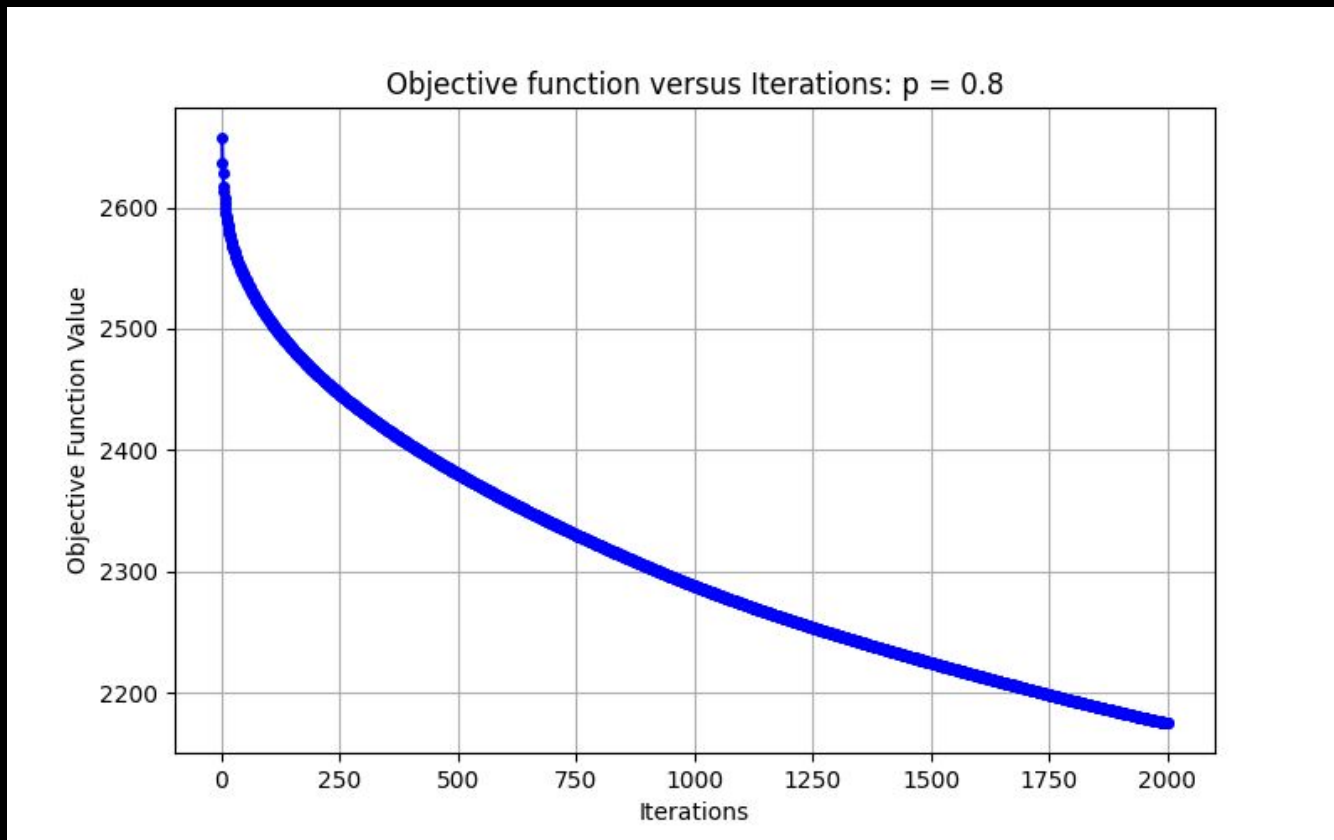
Q3 - Objective Function Plots



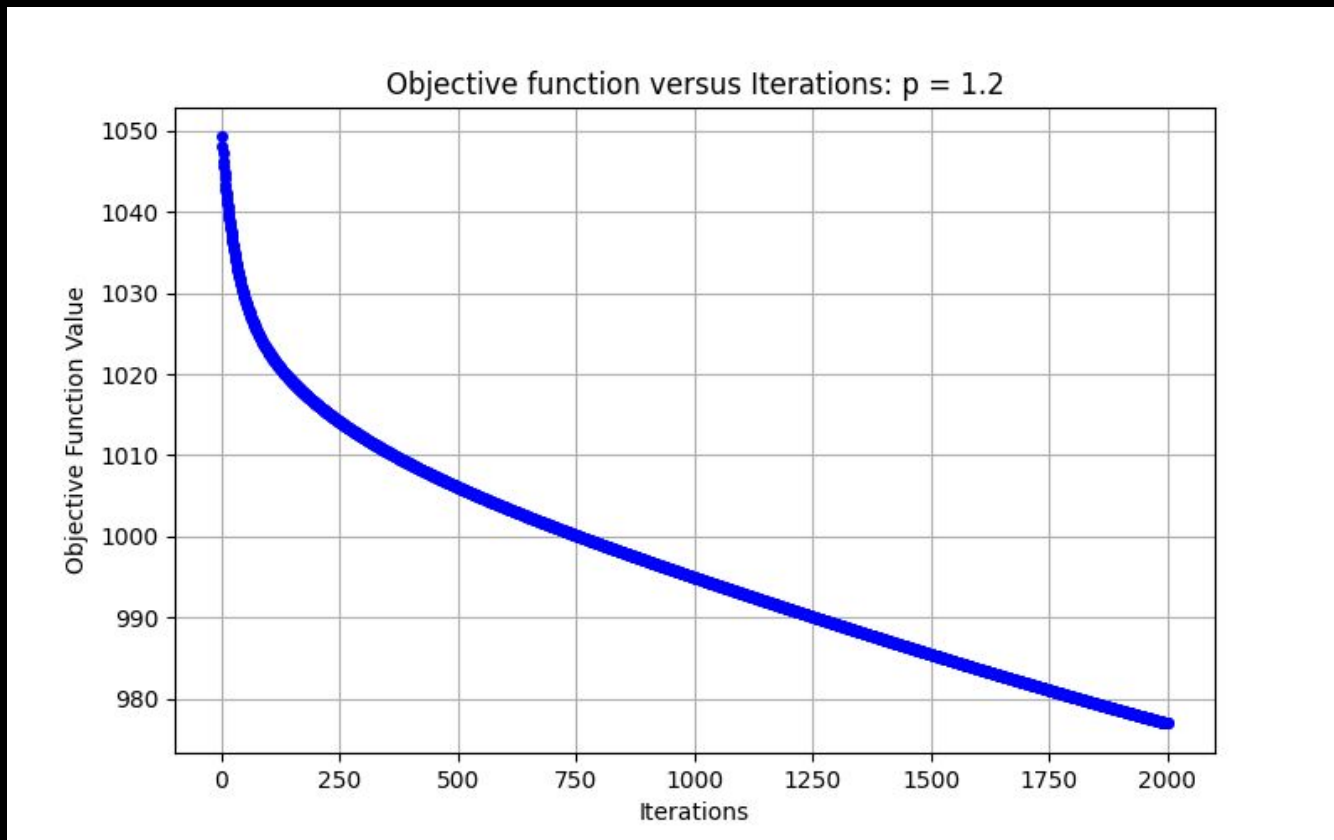
Q4 - RRMSE: Denoising using Dictionary Learning

1. Pre denoising RRMSE : 0.73871
2. Post denoising RRMSE : 0.65972

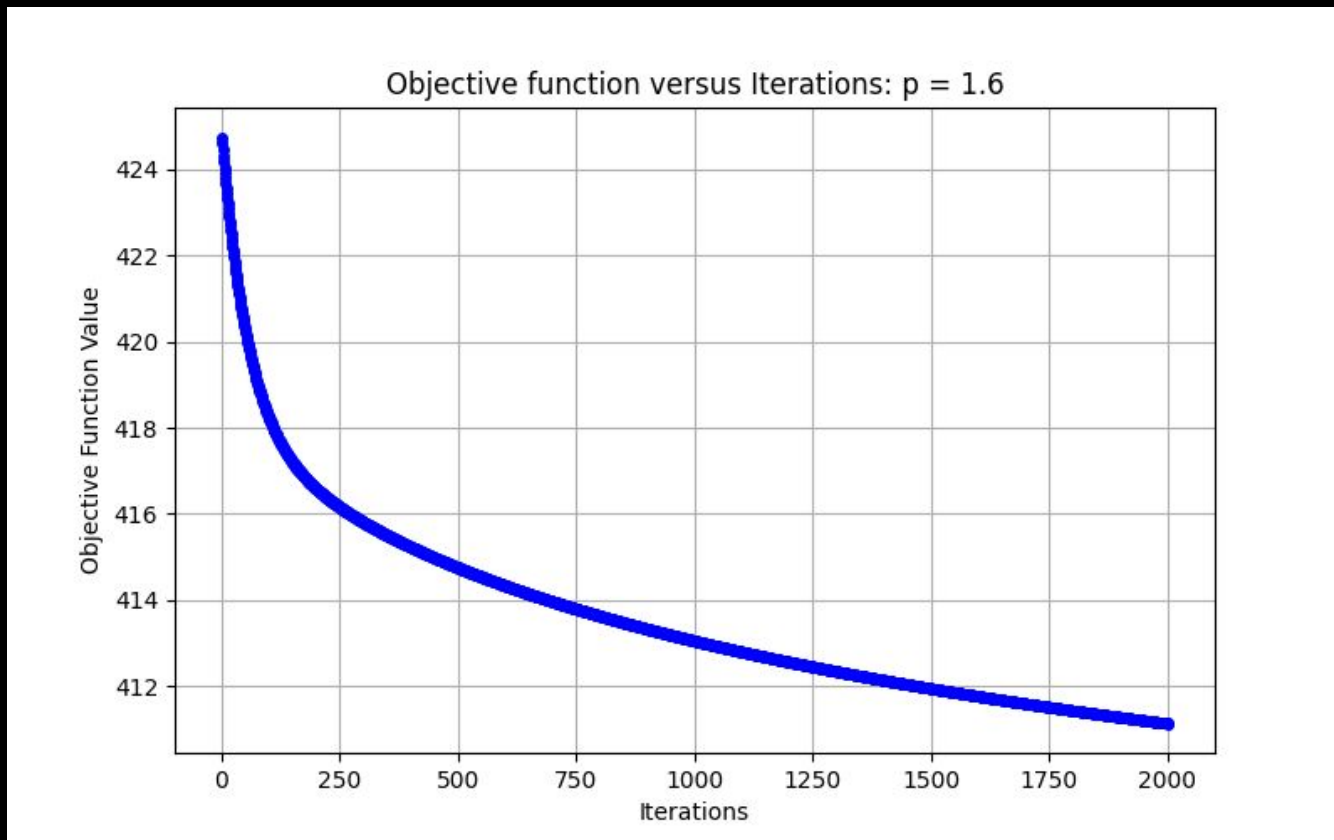
Q4 - Objective Function v/s Iterations(Dictionary Learning)



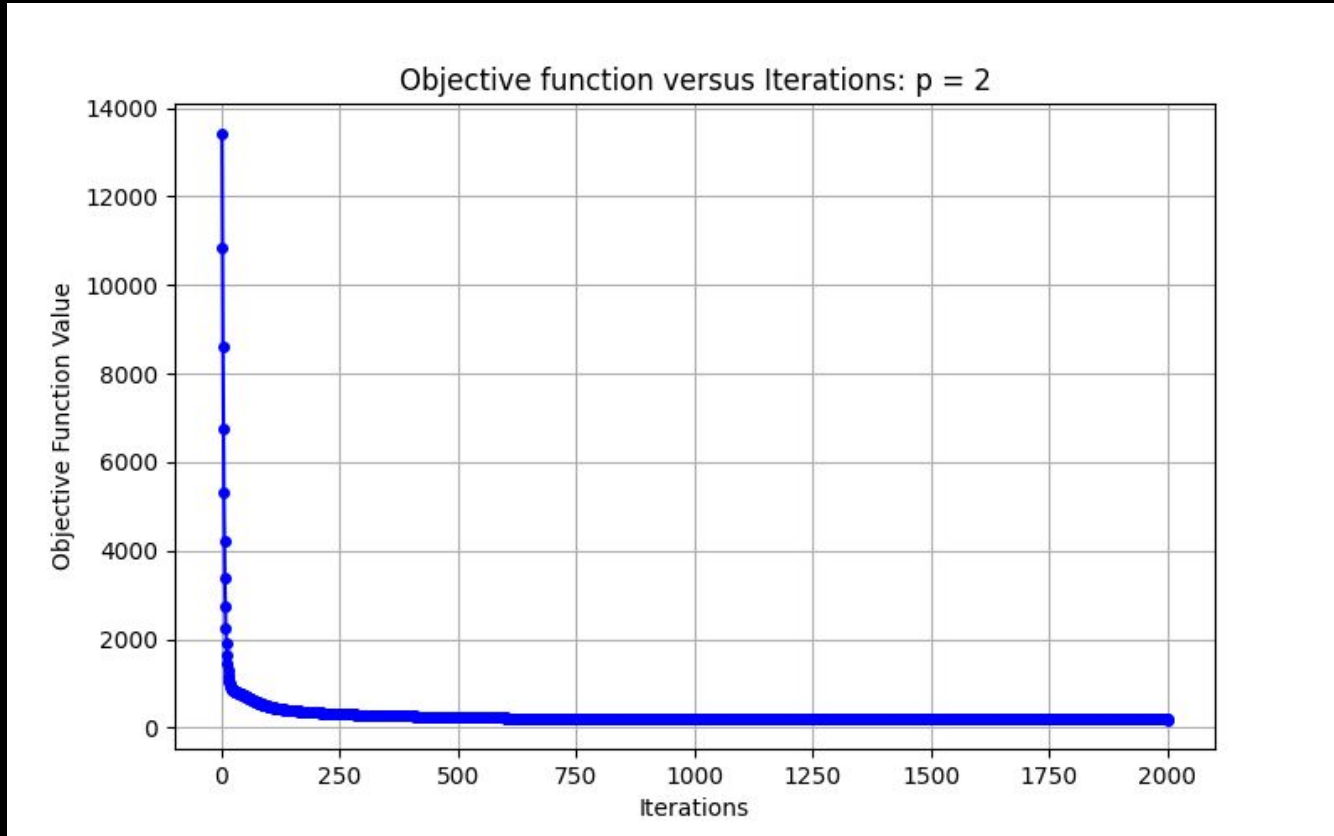
Q4 - Objective Function v/s Iterations(Dictionary Learning)



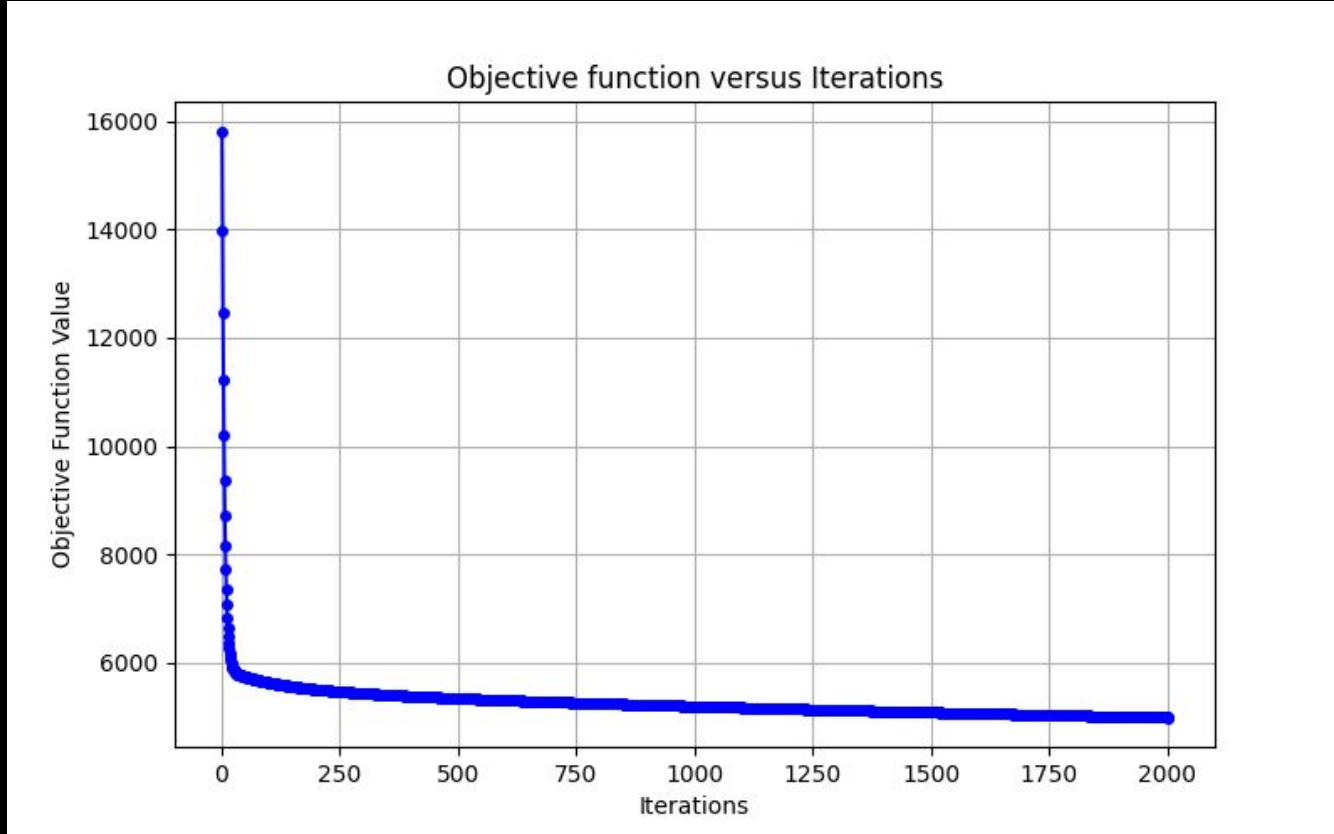
Q4 - Objective Function v/s Iterations(Dictionary Learning)



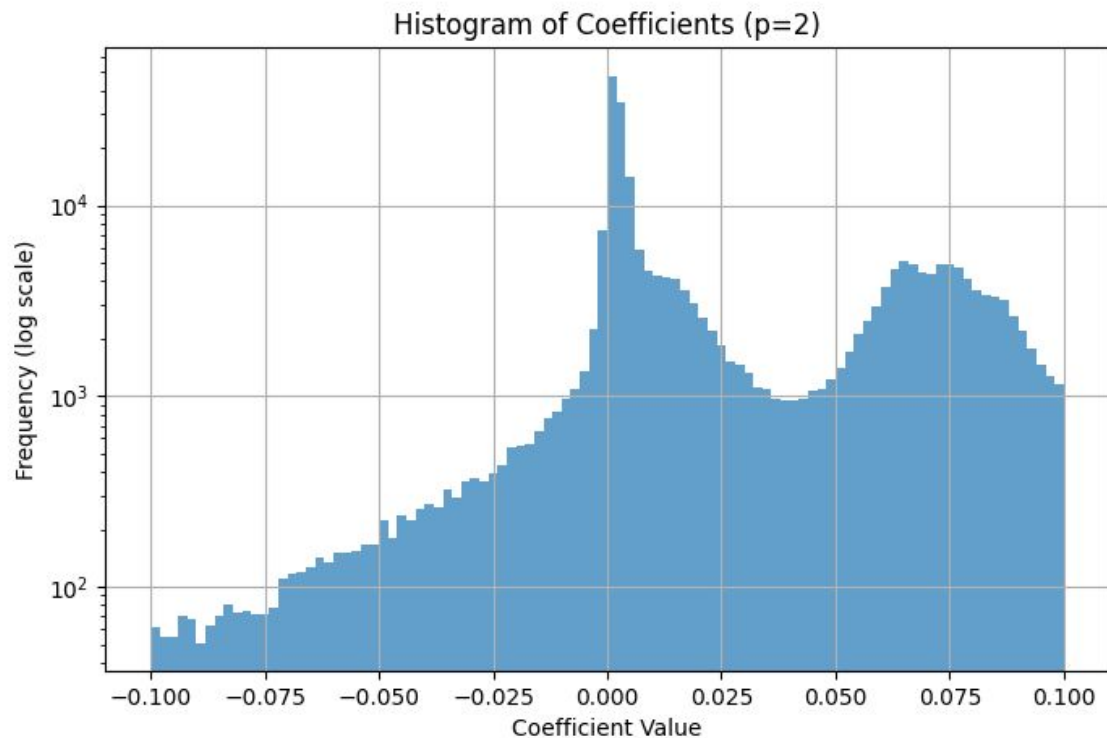
Q4 - Objective Function v/s Iterations(Dictionary Learning)



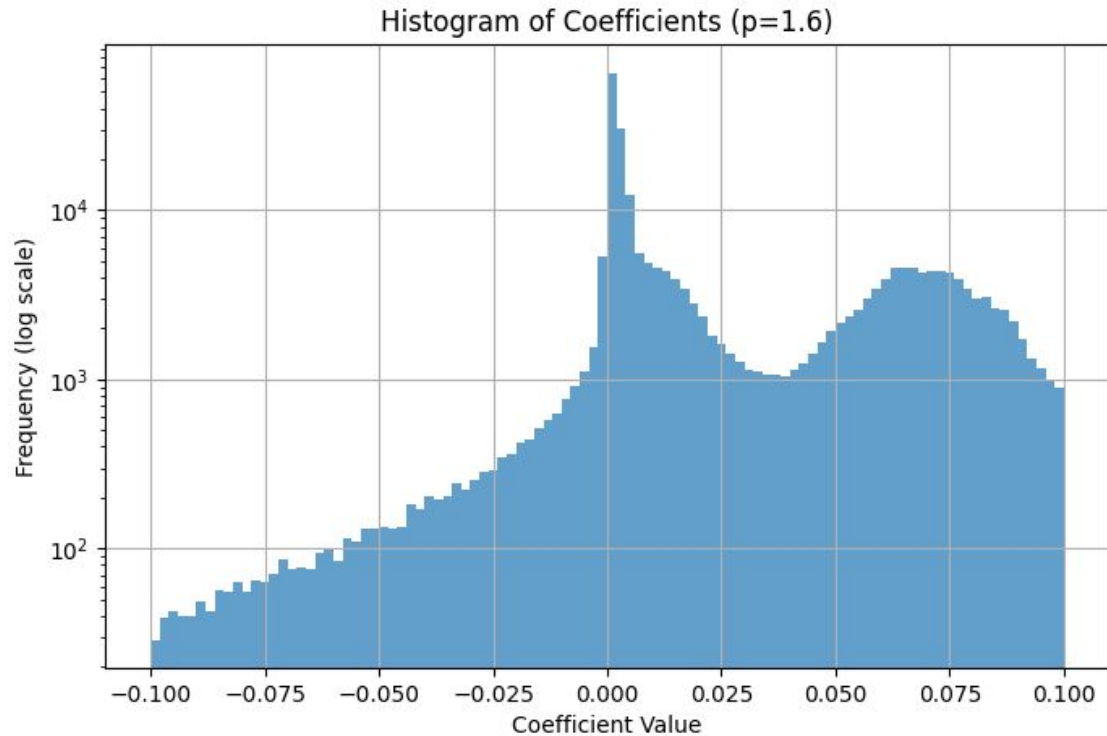
Q4 - Objective Function v/s Iterations(Image Denoising $p=0.8$)



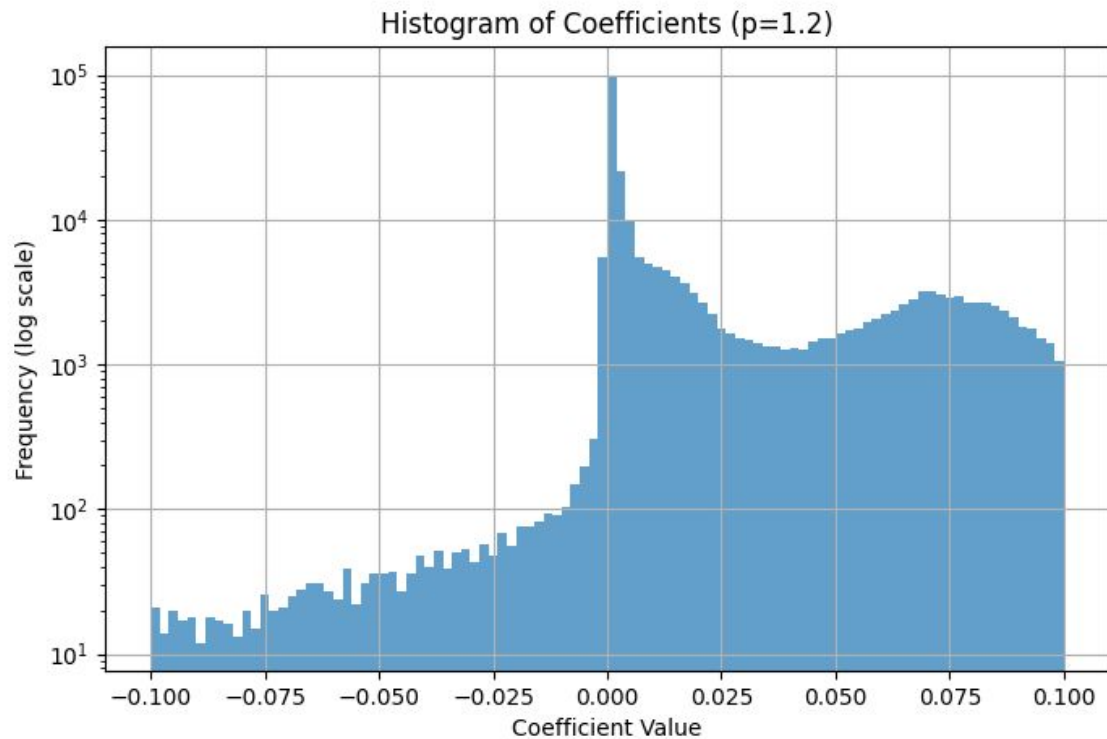
Q4 - Histograms



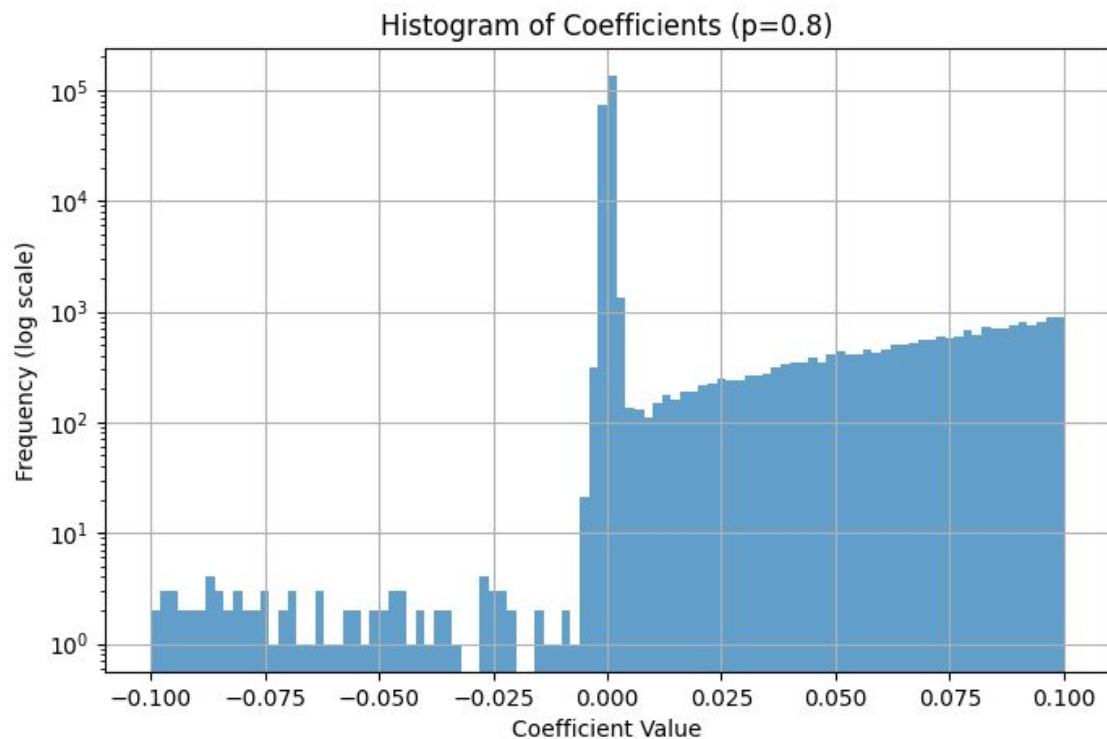
Q4 - Histograms



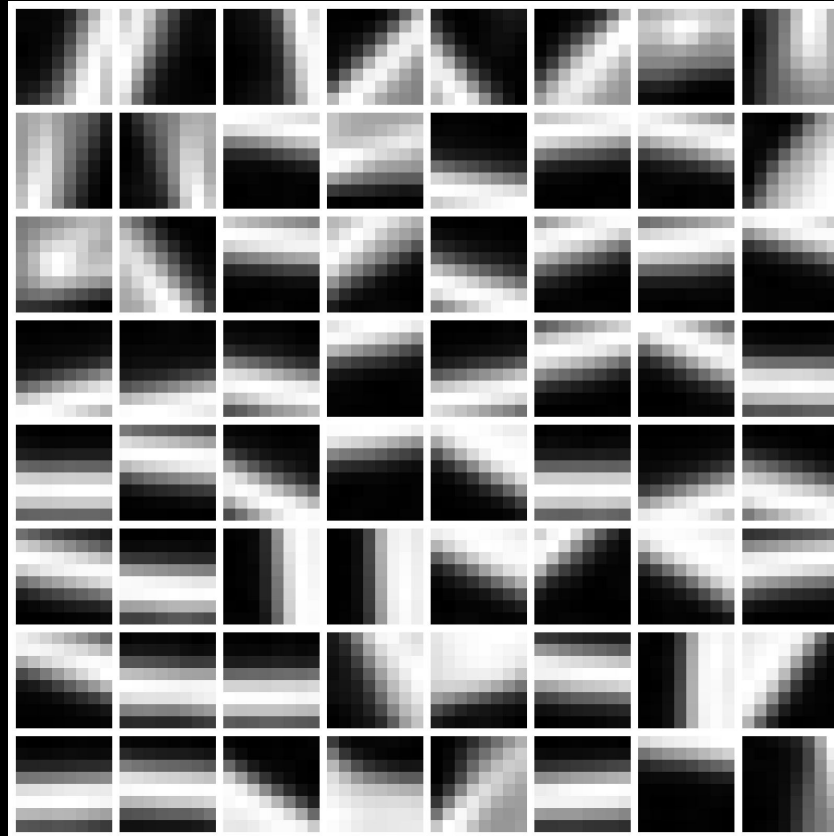
Q4 - Histograms



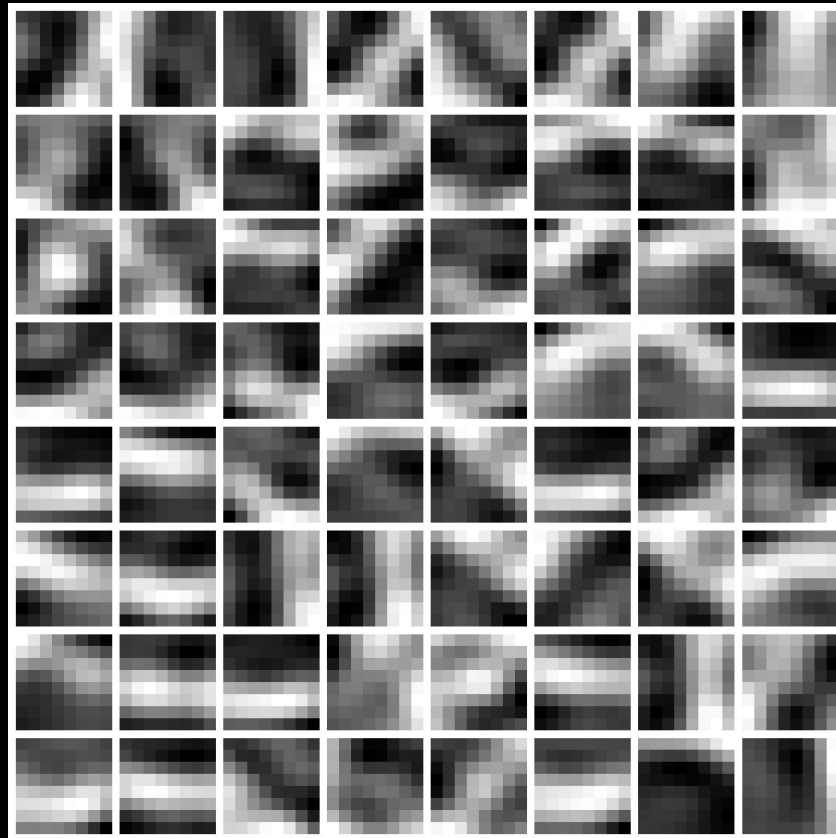
Q4 - Histograms



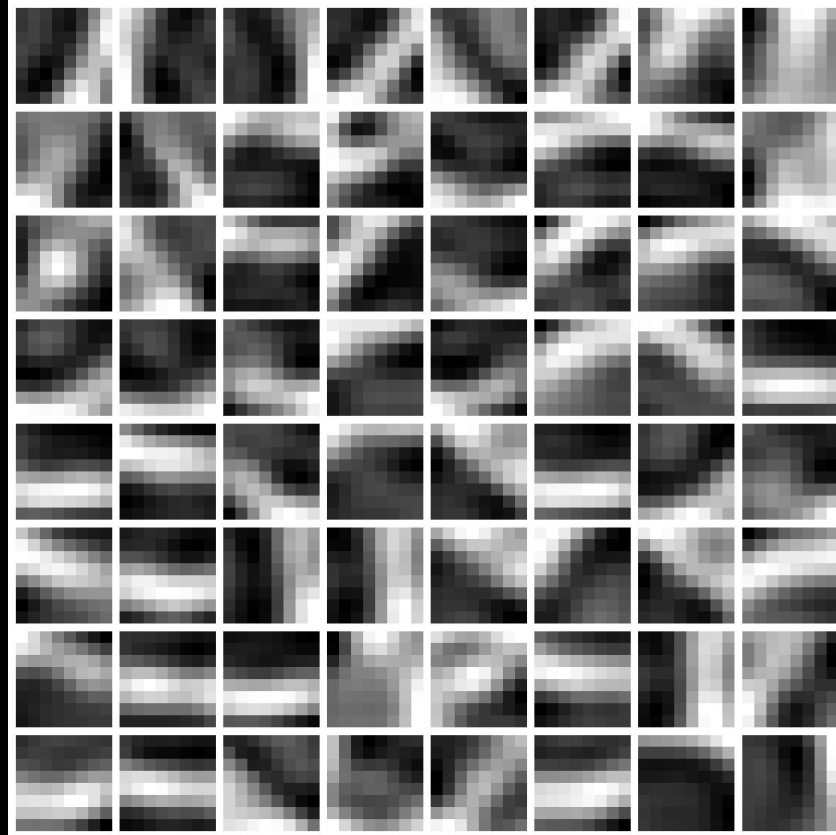
Q4 - Initial Dictionary



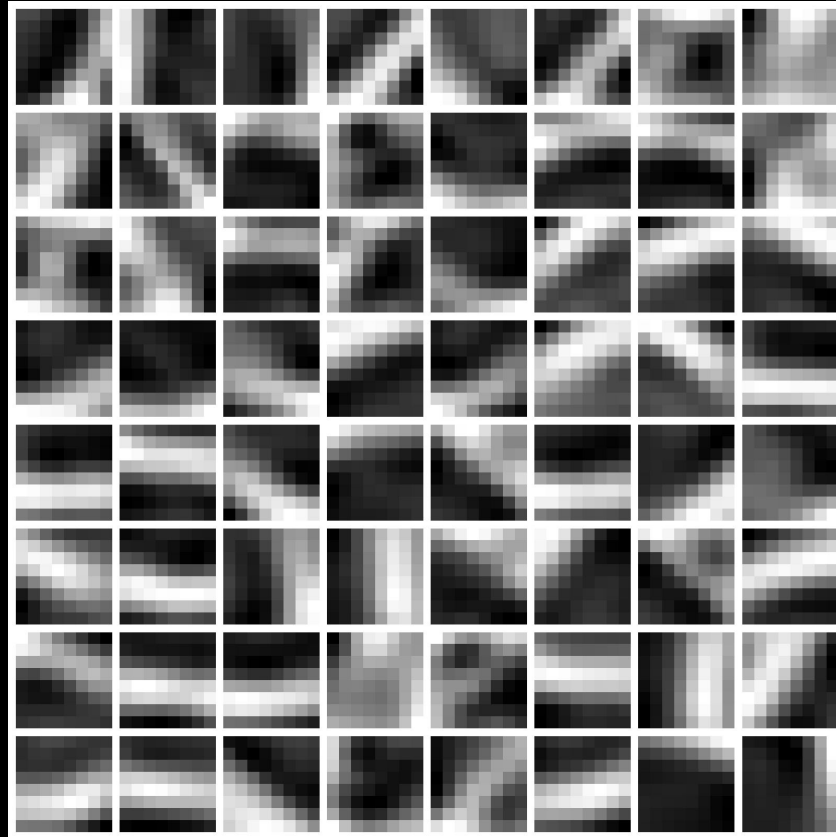
Q4 - Learned Dictionary $p = 2$



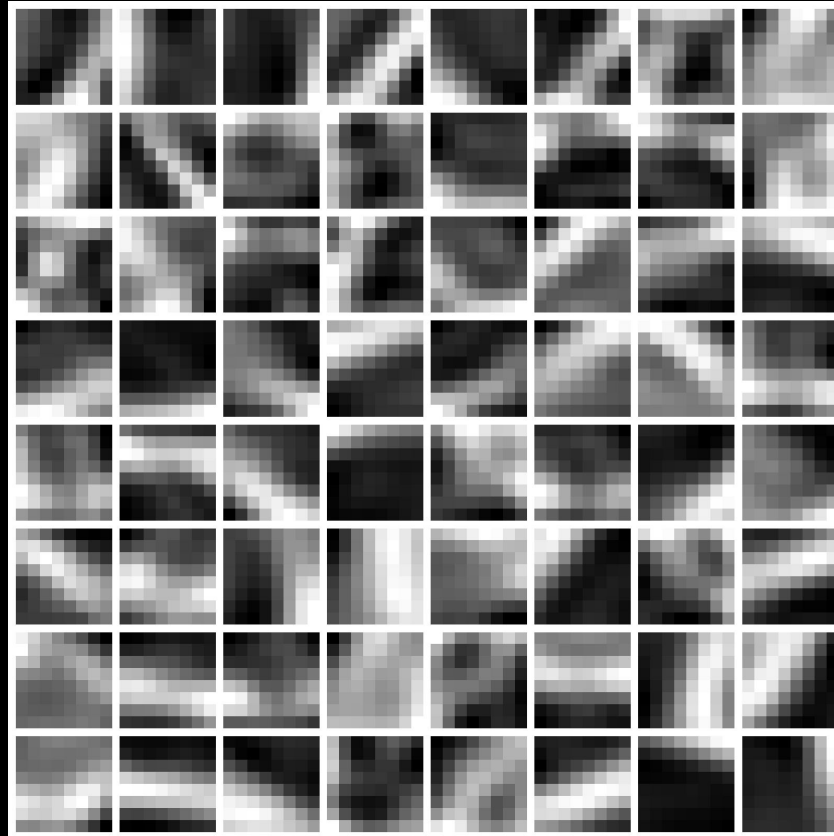
Q4 - Learned Dictionary $p = 1.6$



Q4 - Learned Dictionary $p = 1.2$

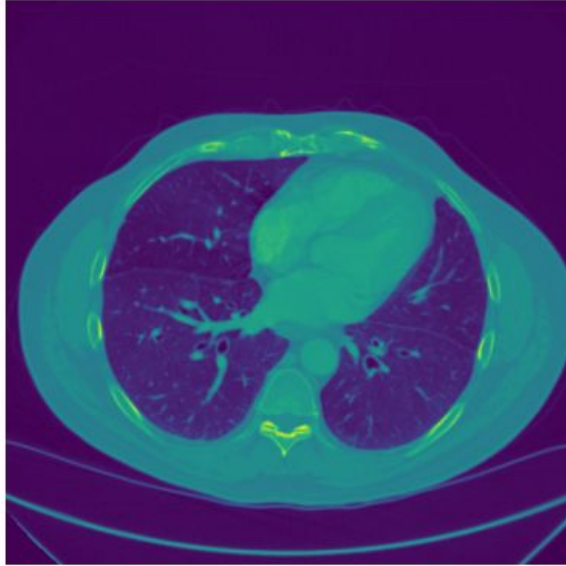


Q4 - Learned Dictionary $p = 0.8$

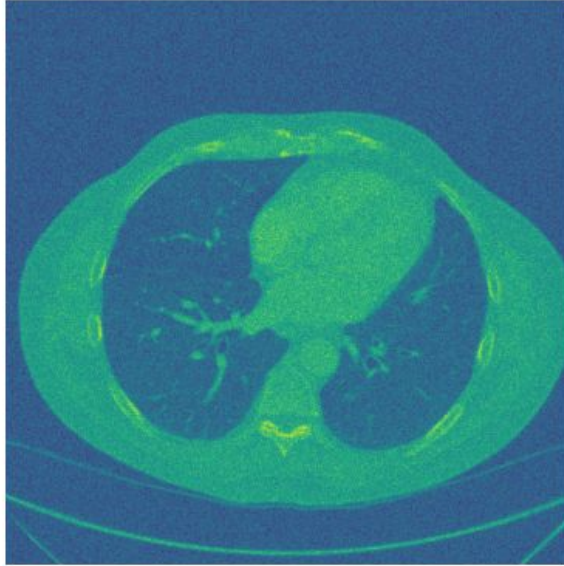


Q4 - Original / Noisy / Denoised

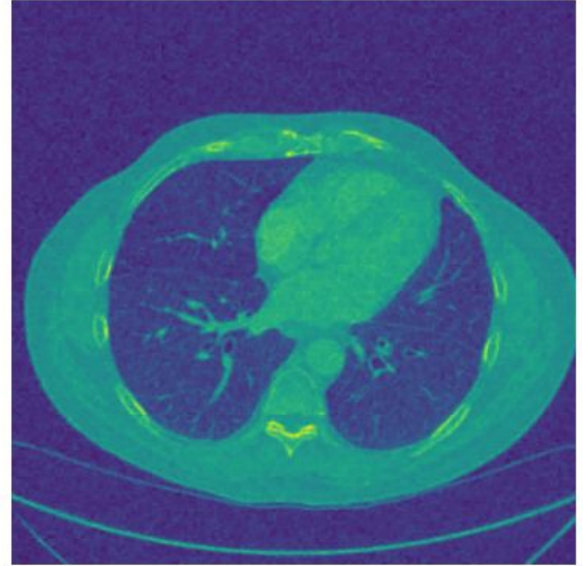
Original Image



Simulated Noisy Image



Denoised Image



Thank You